DBMS MINI-PROJECT REPORT

Comprehensive Student Academic & Activity Ledger

Student Name: _____

Roll Number / Class: _____

Subject: Database Management Systems

Submission Date: _____

Project objective

To design a relational database and graphical interface that stores student profile details, secondary/higher-secondary qualifications, semester-wise SGPA/CGPA, subject and laboratory marks, academic gap years, seminars, and extracurricular/technical activities. The system provides search, progression visualisation, category filters, and spreadsheet export.

1. Tools and implementation

Python is used for application logic, Streamlit provides the browser-based GUI, SQLite stores relational data locally, Pandas prepares tabular exports, and Excel/Google Sheets are used for spreadsheet review. The application creates its database and tables automatically at first launch.

2. Database schema and relationships

Students is the parent table. Qualifications, Gap_Years, Semesters, and Activities reference Students using student_id. Subjects references Semesters using semester_id. A student can have multiple semester records, activities, subjects (through semesters), qualifications, and gap-year records. Foreign keys use ON DELETE CASCADE so linked child records are removed when a student is deleted.

3. Normalisation (3NF rationale)

Student profile data is stored once in Students. Qualifications, semesters, subjects, gap years, and activities are stored in separate tables. Non-key attributes describe the key of their own table, reducing repeated profile and achievement strings. Subject marks depend on the subject record; semester SGPA/CGPA depend on the student's semester record.

4. Constraints

Primary keys identify each row. Roll number is UNIQUE. A student can have only one record per qualification type and one record per semester number. CHECK constraints limit marks and percentages to 0–100, SGPA/CGPA to 0–10, semester number to 1–12, and category/status fields to allowed values. Foreign keys maintain referential integrity.

5. GUI features

The dashboard searches students by name or roll number, displays semester SGPA/CGPA progression as a line chart, lists gap-year markers, and shows subject/lab marks and portfolio activity. Other tabs add students, semester records, subjects/lab marks, activities/seminars, qualifications, and gap years. Activity filters separate Technical/Co-curricular from Extra-curricular and Internal from External seminars.

6. Spreadsheet workflow

Use Export & Database Guide to prepare an Excel workbook containing separate sheets for Students, Qualifications, Semesters, Subjects_Labs, Activities, and Gap_Years. Upload this workbook into Google Sheets using File → Import → Upload. Share the Google Sheets link according to the instructor's access requirement. In this version, spreadsheet transfer is manual; automatic two-way synchronisation is not implemented.

7. How to run

Install Python 3.11 or newer. Extract the project ZIP, open Command Prompt in the extracted folder, run `py -m pip install -r requirements.txt`, then run `py -m streamlit run app.py`. Open the local address shown in the terminal, usually http://localhost:8501.

8. How to make changes safely

Add or update information through the GUI. Use valid roll numbers and avoid duplicates. Enter marks between 0 and 100, SGPA/CGPA between 0 and 10, and a valid semester number. Record lab subjects with subject type Lab and enter internal/external marks separately. Use the correct activity category and seminar type. Export a new workbook after changes and re-import it to Google Sheets. Keep a backup of the database file before making manual database edits.

9. Limitations and future scope

The submitted prototype uses SQLite and manual spreadsheet export. It does not provide live two-way Google Sheets synchronisation, cloud hosting by default, user authentication, or a production-grade audit trail. Future work can use PostgreSQL/MySQL, a validated Google Sheets API integration, role-based access, deployment, backups, and automated tests. The chart displays stored SGPA/CGPA values; it does not calculate grades from marks.

10. Conclusion

The prototype demonstrates relational student record management, integrity constraints, student search, semester progression visualisation, gap-year tracking, portfolio segregation, and spreadsheet export. Demo data is fictional and should be replaced with approved records before any real use.

Important: Insert screenshots of the running GUI and your actual Google Sheets URL before final submission. Do not claim live synchronisation or public hosting unless you have tested it.